

EXPERIMENT 26

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Aim

To write a menu-driven C program using a **switch** statement and nested loops to print three right-aligned patterns. Each pattern is implemented in a separate parameterless function with no return value.

Algorithm

1. Start the program.
2. Display the menu for the three right-aligned patterns.
3. Read the user's choice.
4. Use a switch statement to call the corresponding parameterless void function.
5. In the selected function, input the number of rows n.
6. For every row, first print the required leading spaces and then print stars or natural numbers using nested loops.
7. If the choice is outside 1-3, display an invalid-choice message.
8. Stop.

C Program

```
#include <stdio.h>

void increasingStar(void);
void increasingNumber(void);
void decreasingStar(void);

int main(void) {
    int choice;

    printf("RIGHT-ALIGNED PATTERNS MENU\n");
    printf("1. Right-aligned increasing triangle\n");
    printf("2. Right-aligned increasing number triangle\n");
    printf("3. Right-aligned decreasing triangle\n");
    printf("Enter your choice (1-3): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1: increasingStar(); break;
        case 2: increasingNumber(); break;
        case 3: decreasingStar(); break;
        default: printf("Invalid choice!\n");
    }
    return 0;
}

void increasingStar(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= i; j++) printf("* ");
        printf("\n");
    }
}

void increasingNumber(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= i; j++) printf("%d ", j);
        printf("\n");
    }
}
```

```

void decreasingStar(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = n; i >= 1; i--) {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= i; j++) printf("* ");
        printf("\n");
    }
}

```

Test-Case Table

Test	Choice	n	Expected result
1	1	4	Right-aligned increasing star triangle with 1 to 4 stars
2	2	4	Right-aligned rows: 1; 1 2; 1 2 3; 1 2 3 4
3	3	4	Right-aligned decreasing star triangle with 4 to 1 stars
4	6	-	Invalid choice!

Sample Execution (n = 4)

Choice 1 - Right-aligned increasing triangle

```

*
* *
* * *
* * * *

```

Choice 2 - Right-aligned increasing number triangle

```

1
1 2
1 2 3
1 2 3 4

```

Choice 3 - Right-aligned decreasing triangle

```

* * * *
* * *
* *
*

```

Result

The program successfully uses a switch statement and separate parameterless void functions to generate all three right-aligned patterns. Nested loops control both the leading spaces and the pattern elements, and the prepared test cases verify every menu option plus an invalid choice.